Worksheet 1

Presenting and organising data – DETAILED SOLUTIONS

Badr ABOUFARASSE, PhD. – Academic year 2026–2027

Detailed solution – Exercise 1

The observed population is the **210 surveyed customers**; the statistical unit is **one customer**. The variable is the **maximum acceptable price**. It is a **continuous quantitative variable**. Satisfaction (satisfied/not satisfied) is a **qualitative variable**. A bar chart or pie chart is suitable. A mean and standard deviation are not appropriate for these qualitative categories.

Detailed solution – Exercise 2

$$N = 4 + 12 + 32 + 32 + 60 + 35 + 25 + 5 + 5 = 210.$$

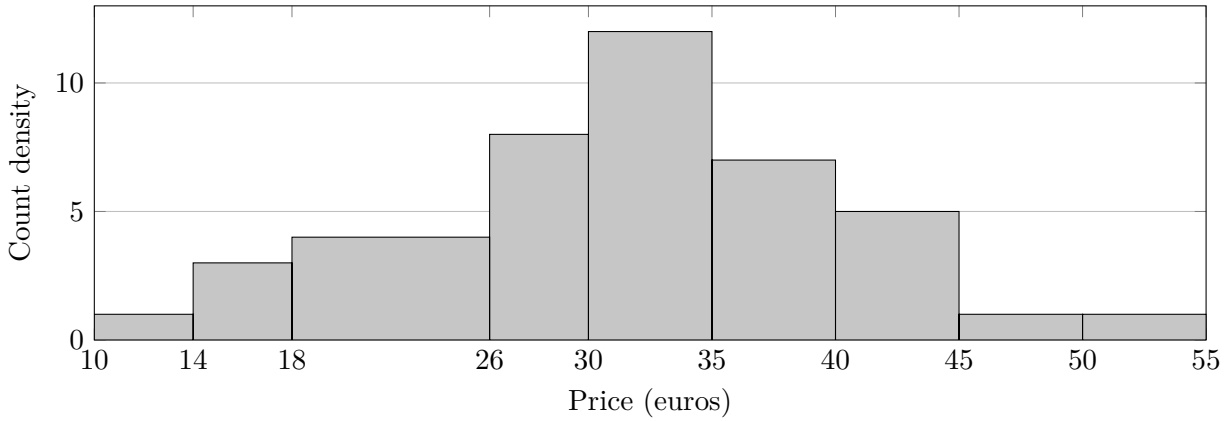
For each class, $f_i = n_i/210$ and $d_i = n_i/a_i$.

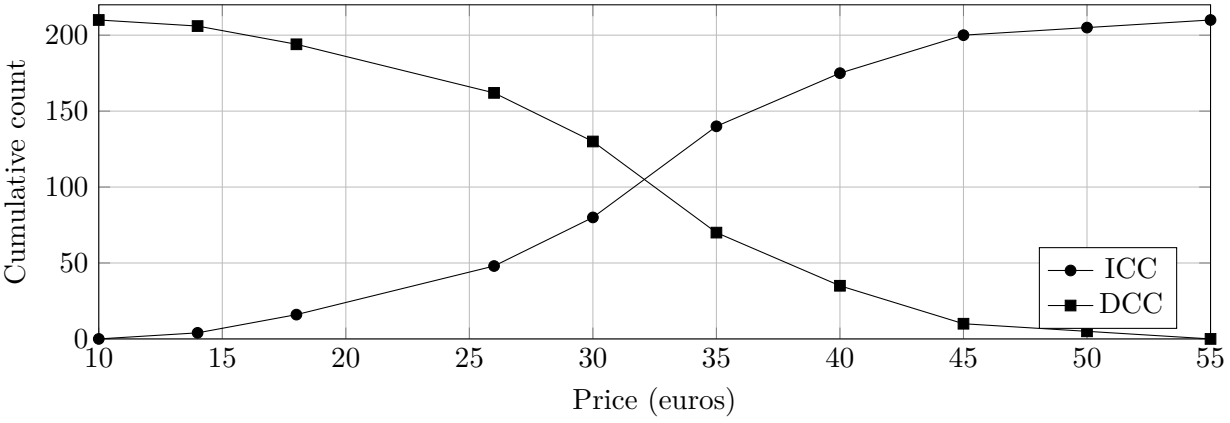
Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5
a_i	4	4	8	4	5	5	5	5	5
f_i (%)	1.90	5.71	15.24	15.24	28.57	16.67	11.90	2.38	2.38
d_i	1	3	4	8	12	7	5	1	1
ICC	4	16	48	80	140	175	200	205	210
DCC	210	206	194	162	130	70	35	10	5
ICRF (%)	1.90	7.62	22.86	38.10	66.67	83.33	95.24	97.62	100
DCRF (%)	100	98.10	92.38	77.14	61.90	33.33	16.67	4.76	2.38

Example:

$$f_{[18,26[} = 32/210 = 15.24\%, \quad ICC_{[26,30[} = 4 + 12 + 32 + 32 = 80.$$

With unequal class widths, rectangle *area* must represent the count, so height is the density n_i/a_i.



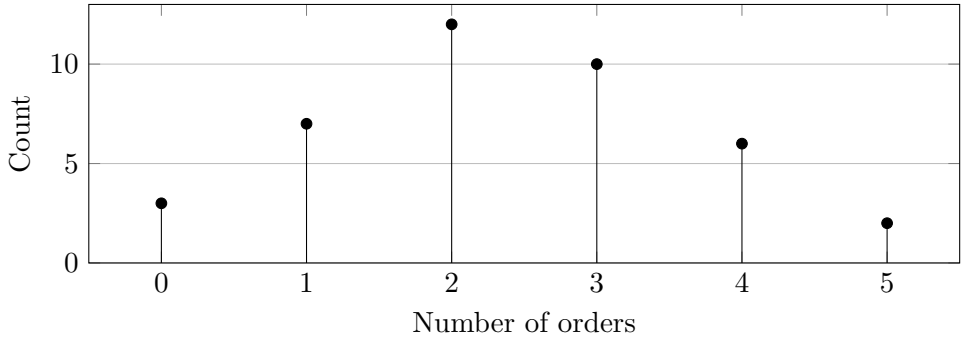


Less than 35 euros: $140/210 = 66.67\%$. At least 35 euros: $70/210 = 33.33\%$.

Detailed solution – Exercise 3

x_i	0	1	2	3	4	5
n_i	3	7	12	10	6	2
f_i (%)	7.5	17.5	30	25	15	5
Cumulative (%)	7.5	25	55	80	95	100

For example, $f_2 = 12/40 = 30\%$ and $F(3) = 32/40 = 80\%$.



Thus, **80% of customers placed at most 3 orders**.